

Instrumentation and Control

CSE 325

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Measurement of Pressure

Pressure

Pressure is the **normal force per unit area** exerted by a **fluid (liquid or gas)** on a surface (usually the walls of the container).

Pressure acts **perpendicular (normal)** to the surface.

Pressure is **not a vector quantity**; it is a **scalar**, meaning it has magnitude only and no direction

Types of Pressure

There are three commonly used types of pressure:

1. Absolute Pressure

Absolute pressure is the actual pressure measured relative to a **perfect vacuum**, where pressure is zero.

2. Gauge Pressure

Gauge pressure is the difference between the **measured pressure** and the **atmospheric (ambient) pressure**.

3. Differential Pressure

Differential pressure is the difference between **two pressure points**, where one point is taken as a reference. Even if both pressures change, the difference between them is what matters.

Electro-Mechanical Pressure Measurement

These instruments use **both mechanical and electrical methods** to measure pressure.

- **Mechanical part** → senses (detects) the pressure
- **Electrical part** → shows or records the pressure

Main Components

These instruments need **two main parts**:

1.Sensing Element

2.Electrical Transducer

Because of this combination, they are called **secondary transducers**.

A transducer is a device that converts energy from one form into another

Sensing Element

- known as a **force-summing device**
- Its job is to **detect pressure**
- It converts pressure into **mechanical movement (displacement)**

In simple terms:

Pressure → turns into movement

Electrical Transducer

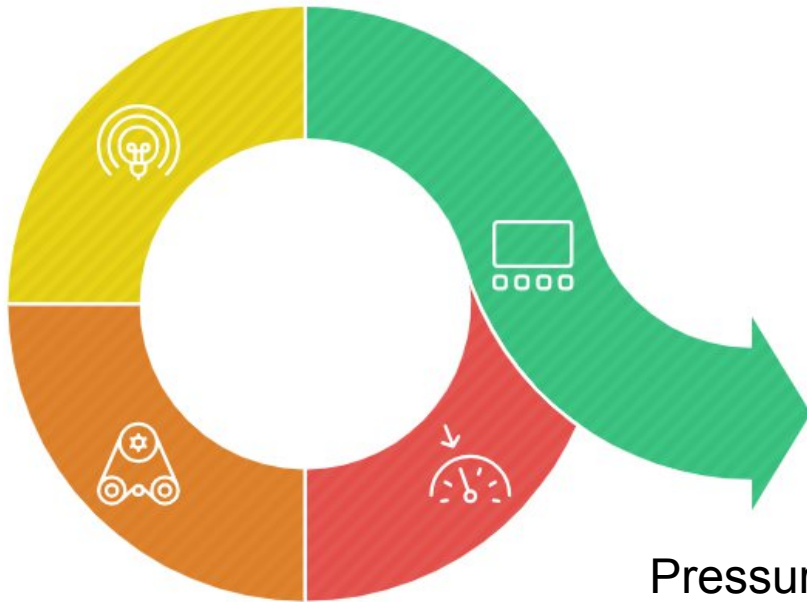
- It receives the **movement from the sensing element**
- Converts that movement into an **electrical signal**
- This signal is then used for **measurement or display**.

In simple terms:

Movement → turns into electrical signal

Electro-Mechanical Pressure Measurement

Pressure Measurement Cycle



Pressure → Mechanical Movement → Electrical Signal → Display

1

Pressure Applied

Pressure is exerted on the sensor.

2

Mechanical Movement

Pressure causes mechanical movement.

3

Electrical Signal

Movement is converted to an electrical signal.

4

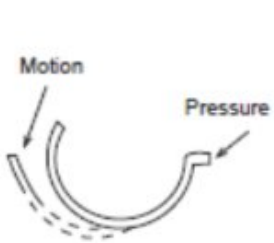
Display

Signal is displayed for measurement.

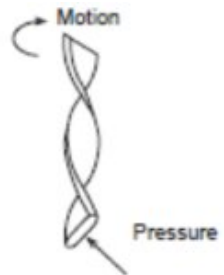
Electro-Mechanical Pressure Measurement

Sensing element:

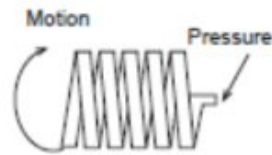
- must be made of an **elastic material** and must **operate** within its **elastic limit**
- The stresses produced in the sensing element establishes equilibrium with **the pressure to be measured by a displacement of the element**
- The **choice and design** of the type of sensing element **depends on the magnitude of pressure to be measured**.



C-shaped Bourdon tube;



twisted Bourdon tube;



helical Bourdon tube;

Secondary Transducers

various types of transducers are used

- ☁ Resistive,
- ☁ Inductive
- ☁ Capacitive
- ☁ Piezoelectric

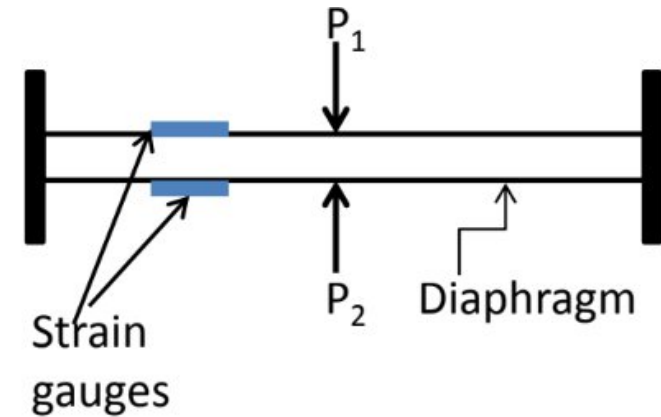
Resistive Pressure Transducer

Diaphragm-Strain Gauge pressure sensor

Sensing element: **diaphragm**

Transducer: **strain gauge** are attached to a diaphragm

e.g. **Two strain gauges** are attached on **opposite sides of a diaphragm** for **differential pressure measurement**



Working Principle

The process works step by step:

Differential Pressure → **Diaphragm Deflection** → **Strain Difference**
→ **Electrical Output**

- When pressure is applied, it causes the **diaphragm to bend (deflect)**
- This bending creates a **difference in strain** in two strain gauges
- The strain difference is converted into an **electrical signal**

Capacitive Pressure Transducers

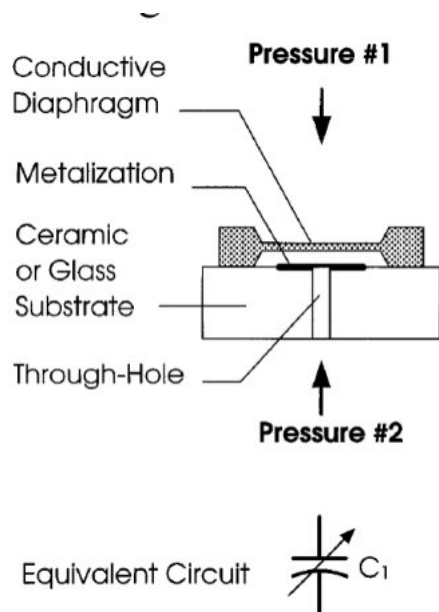
Highly accurate pressure sensors of today use capacitive detection technique

Ex: Diaphragm – Capacitive pressure sensor

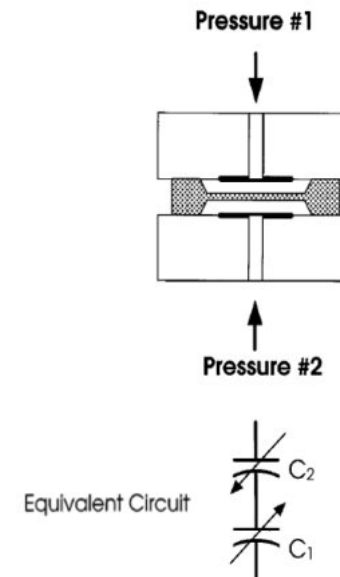
- Sensing element: diaphragm
- Transducer: Capacitive

Pressure diaphragm movement change in distance between plates

Option 1: single capacitor design



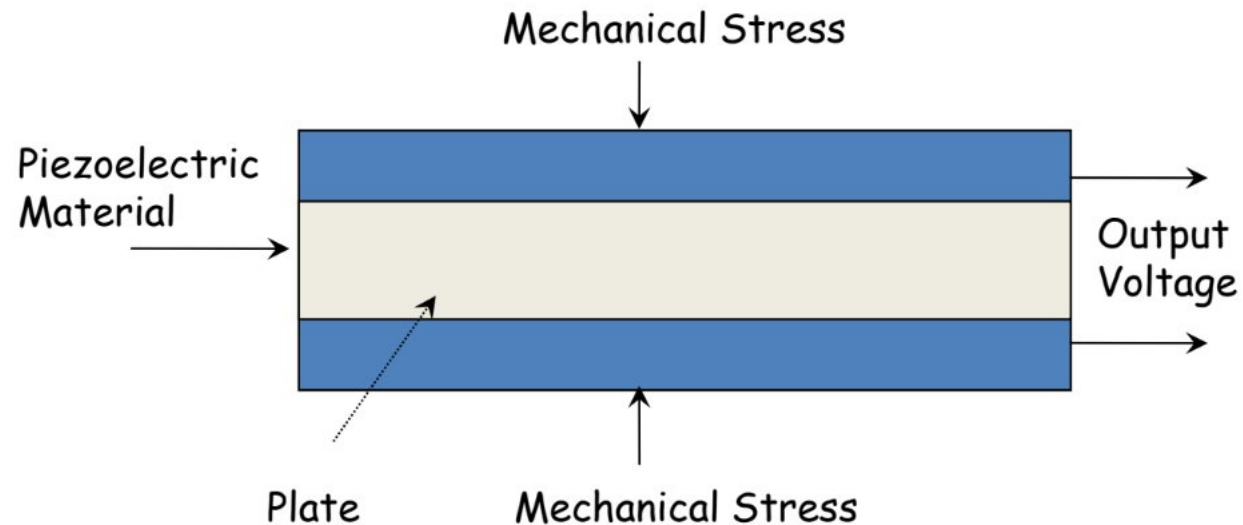
Option 2: Differential capacitor design



Piezoelectric Pressure Transducers

This is a primary pressure transducer; **NO force summing device** is needed

- Piezoelectric crystal (made from quartz): **converts mechanical stress applied to it into voltage.**
- **Pressure** exerted on the crystal surface is **proportional** to the **voltage produced by the crystal.**
- No voltage supply is required. It produces voltage.
- Suitable for fast-changing pressure measurement.



Measurement of Flow

Ultrasonic Flowmeter

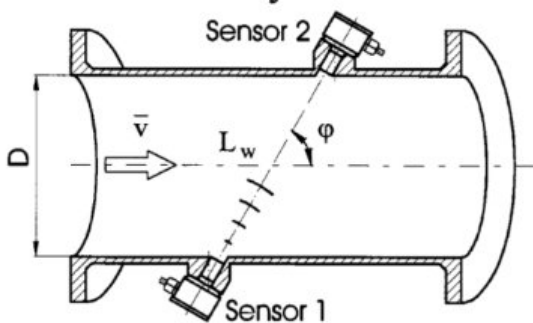
Various types of ultrasonic flowmeters are there for flow/discharge measurement

- Most widely used type is the Transit Time type Transit time type

Operating principle:

- makes use of the difference in the time for an ultrasonic pulse to travel a fixed distance, first against the flow and then in the direction of flow.
- the propagation velocity of an acoustic wave and the flow velocity are summed vectorially.

The two transmit times (t_{12} and t_{21}) of acoustic pulses are given by:



$$t_{12} = \frac{L_w}{c + v_a \cos \phi} \quad \text{and} \quad t_{21} = \frac{L_w}{c - v_a \cos \phi}$$

where L_w = Distance in the fluid between the two transducers

c = Speed of sound at the operating conditions

ϕ = Angle between the axis of the conduit and the acoustic path

$\bar{v}_a$ = Axial flow velocity averaged along the distance L_w

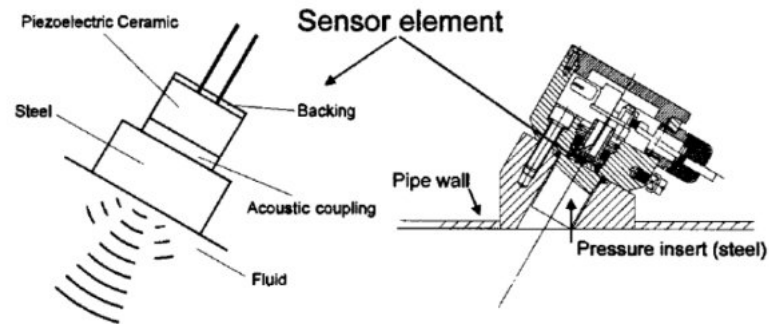
Measurement of Flow

The mean axial fluid flow velocity can be found as:

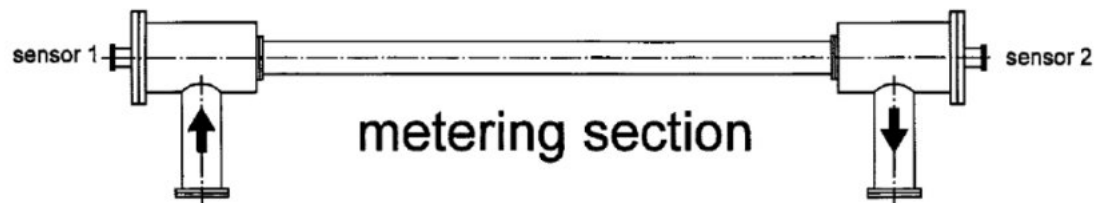
$$\bar{v}_a = \frac{L_w}{2 \cos \varphi} \left(\frac{1}{t_{21}} - \frac{1}{t_{12}} \right) = \frac{D}{2 \cos \varphi \sin \varphi} \left(\frac{1}{t_{21}} - \frac{1}{t_{12}} \right)$$

Sensors:

Transmitter and receiver both are made of piezoelectric crystals

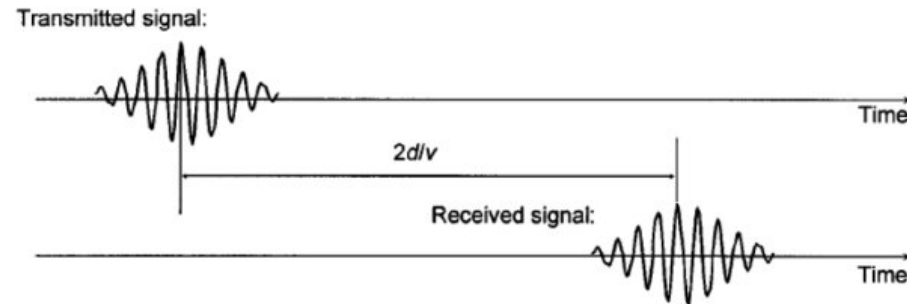
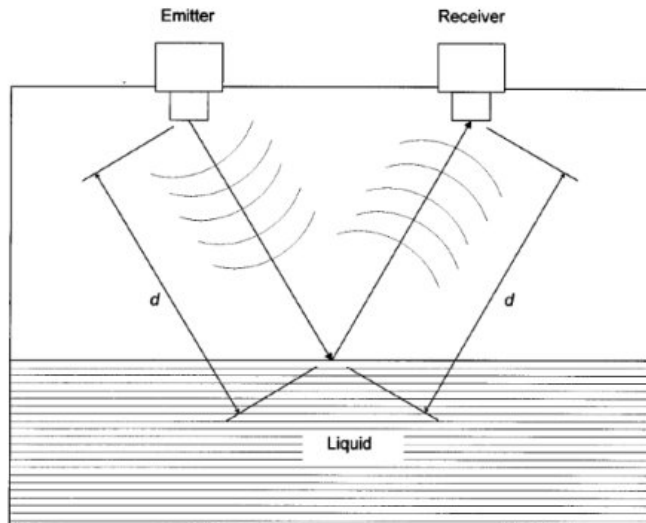


Axial metering:



Measurement of Liquid Levels

Time-of-Flight Method



The measuring system evaluates the time-of-flight t of the signal:

$$t = \frac{2d}{v}$$

Where v is the propagation velocity of the waves

The waves can be:

- a. Ultrasonic: longitudinal acoustic waves; same/separate piezoelectric crystals can be used as emitter and receiver
- b. Electro-magnetic: could be microwave, laser; wave generation and detection are performed by separate modules.

Measurement of Liquid Levels

Time-of-Flight Method

ultrasonic method:

Transit time is a measure of distance between top liquid surface and the ultrasonic transducer

$$t \propto d \propto (H - h)$$

d is the distance between top surface and transducer, H is fixed distance of transducer from the liquid bottom, h is the liquid height

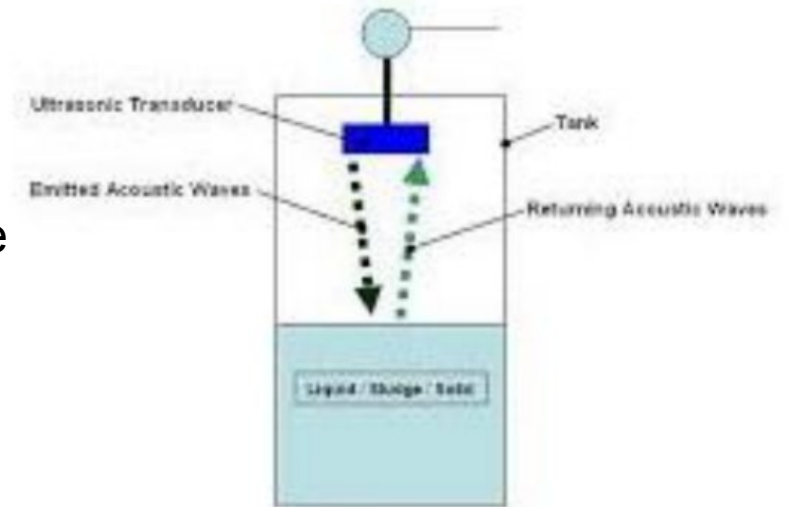


Fig.1.1 – Working of Ultrasonic Level Sensor